CrewSim Data Dictionary

> Each section below corresponds to a dataset file expected by the simulator. The sample CSVs in this folder contain 20 rows each so your colleague can see the exact format, column order, and value patterns required. `used_in_code` indicates whether the field is actively consumed by simulation, calibration, or validation logic (Yes), or only declared in the schema for completeness / future use (No).

D1 — `d1_crew_master.csv`

Crew roster master list. One row per crew member, containing identity, base assignment, fleet qualification, seniority, and employment status.

Field	Type	Description	used_in_code
crew_id	string	Opaque surrogate key identifying the crew member (e.g. `P00001`). Must NOT be a real name or employee number (Law 25).	Yes
base	string	Home base IATA code (e.g. `YUL`, `YYZ`, `YVR`).	Yes
crew_group	string	Crew group: `PILOT` or `CABIN`. Used to filter the population scope.	Yes
position	string	Seat position: `CA` (Captain) or `FO` (First Officer) for pilots.	Yes
fleet	string	Fleet type the crew member is qualified on (e.g. `A321-200`, `A330-200`).	Yes
fleet_family	string	Fleet family grouping (e.g. `A321`, `A330`). Broader than fleet.	No
qual_valid_from	date	Start date of the current type-rating qualification period.	No
qual_valid_to	date	Expiry date of the current type-rating qualification.	No
seniority_number	integer	Company-wide seniority rank (1 = most senior). Drives drafting order.	Yes
employment_status	string	Current status: `ACTIVE`, `INITIAL_TRAINING`, `LEAVE`, etc. Only `ACTIVE` crew enter the simulation.	Yes
hire_date	date	Date the crew member was hired.	No
termination_date	date	Date of termination, if applicable. Empty for active crew.	No

D2 — `d2_rosters.csv`

Published roster blocks. One row per calendar block (pairing, day off, vacation, reserve, training, simulator) per crew member, covering the full data window.

Field	Type	Description	used_in_code
crew_id	string	Foreign key to D1.	Yes
base	string	Base for this roster period.	Yes
fleet	string	Fleet for this roster period.	Yes
position	string	Position for this roster period.	Yes
block_type	string	Type of block: `PAIRING`, `DAY_OFF`, `VACATION`, `RESERVE`, `TRAINING`, `SIMULATOR`, `REST`.	Yes
start_utc	datetime	Block start in UTC (ISO 8601, minute precision).	Yes
end_utc	datetime	Block end in UTC.	Yes
start_local	datetime	Block start in local (base) time. Used for legality duty-window checks.	Yes
end_local	datetime	Block end in local time.	No
pairing_id	string	Foreign key to D3 when block_type is `PAIRING`. Empty otherwise.	Yes
reserve_call_type	string	`SC` (short-call) or `LC` (long-call) when block_type is `RESERVE`. Empty otherwise.	Yes

D3 — `d3_pairings.csv`

Pairing header records. One row per pairing (a multi-day trip sequence), with summary attributes.

Field	Type	Description	used_in_code
pairing_id	string	Unique pairing identifier (e.g. `YULA321-000001`). Foreign key target for D2, D3 legs, D4, D5, D6, D7, D10.	Yes
base	string	Originating base for the pairing.	Yes
fleet	string	Fleet type assigned to the pairing.	Yes
position_complement	string	Required crew complement per position (e.g. `CA:1,FO:1,FA:5`).	No
duty_periods	integer	Number of duty periods (work days) in the pairing.	No
credit_hours	float	Total credited flight hours for pay purposes.	No
layover_stations	string	Pipe-separated list of layover station codes (e.g. `TLS|VCE|GCM`).	No
start_date	date	Date the pairing begins.	No
days	integer	Total calendar days spanned by the pairing.	No

D3 — d3_pairing_legs.csv

Individual flight legs within each pairing. One row per leg, with timing and operational details.

Field	Type	Description	used_in_code
pairing_id	string	Foreign key to D3 pairings header.	Yes
duty_period	integer	Duty period number within the pairing (1-based).	Yes
leg_seq	integer	Leg sequence within the duty period (1-based).	No
flight_no	string	Flight number (e.g. TS238).	No
origin	string	Departure station IATA code.	No
destination	string	Arrival station IATA code.	No
report_local	datetime	Crew report time in local time (includes pre-flight duty).	Yes
dep_local	datetime	Scheduled departure in local time.	No
arr_local	datetime	Scheduled arrival in local time.	No
release_local	datetime	Crew release time in local time (includes post-flight duty).	Yes
block_hours	float	Block time for this leg in hours (chock-to-chock).	Yes

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D4 — d4_absences.csv

Crew absence events (sick calls, injuries, family leave). One row per absence occurrence. This is the primary input for calibrating the sick-call arrival model (NHPP).

Field	Type	Description	used_in_code
absence_id	string	Unique absence identifier (e.g. A000001).	Yes
crew_id	string	Foreign key to D1.	Yes
notification_ts	datetime	Timestamp when the absence was reported (drives notification-to-departure lead time).	Yes
absence_start	date	First day of absence. Key date for NHPP arrival counting and regime joining.	Yes
absence_end	date	Last day of absence.	Yes
duration_days	integer	Total absence duration in calendar days. Used to fit the absence-duration distribution.	Yes
absence_type	string	Category: SICK, INJURY, FAMILY.	Yes
affected_pairing_id	string	The pairing that was affected (uncovered) by this absence. Foreign key to D3.	Yes
base	string	Crew member's base at time of absence.	Yes

Field	Type	Description	used_in_code
fleet	string	Crew member's fleet at time of absence.	Yes
position	string	Crew member's position at time of absence.	Yes

D5 — `d5_reserve_callouts.csv`

Reserve activation records. One row per reserve crew member contacted for a coverage assignment.

Field	Type	Description	used_in_code
callout_id	string	Unique callout identifier (e.g. `R000001`).	No
crew_id	string	Foreign key to D1 — the reserve crew member contacted.	Yes
call_ts	datetime	Timestamp when the reserve was called.	Yes
report_ts	datetime	Timestamp when the reserve reported for duty.	No
assigned_pairing_id	string	The pairing the reserve was assigned to cover. Foreign key to D3.	Yes
outcome	string	Result of the callout: `assigned`, `released`, `no-show`, `declined`.	Yes
base	string	Reserve's base.	Yes
fleet	string	Reserve's fleet.	Yes

D6 — `d6_schedule_changes.csv`

Schedule modification records. One row per roster change event, capturing drafts (`SHORT_NOTICE_COVER`) and cascade effects (`CASCADE_UNCOVERED`). Used to reconstruct historical drafting and validate the cascade model.

Field	Type	Description	used_in_code
change_id	string	Unique change identifier (e.g. `S000001`).	Yes
crew_id	string	Foreign key to D1 — the affected crew member.	Yes
change_ts	datetime	Timestamp of the schedule change.	Yes
old_block_type	string	Block type before the change (e.g. `DAY_OFF`, `PAIRING`).	Yes
new_block_type	string	Block type after the change (e.g. `PAIRING`, `OPEN`).	Yes
new_pairing_id	string	The pairing assigned in the change, if applicable. Foreign key to D3.	Yes
reason_code	string	Reason: `SHORT_NOTICE_COVER` (draft), `CASCADE_UNCOVERED`, `VOLUNTARY_SWAP`, etc.	Yes

Field	Type	Description	used_in_code
initiator	string	Who initiated the change: `IOC` (operations control), `CREW`, or empty.	Yes

D7 — `d7_pay_register.csv`

Payroll records related to reserve utilization and drafting premiums. Used for backtest cost validation (gate V6).

Field	Type	Description	used_in_code
crew_id	string	Foreign key to D1.	Yes
pay_period	string	Bid period in `YYYY-MM` format (e.g. `2023-08`).	Yes
pay_code	string	Payment category: `RESERVE_UTIL`, `DRAFT_PREM`, `POSITIONING`, etc.	Yes
amount_cad	float	Payment amount in Canadian dollars.	Yes
duty_date	date	The specific duty date this payment relates to.	No
linked_pairing_id	string	The pairing this payment is linked to. Foreign key to D3.	Yes

D8 — `d8_flight_movements.csv`

Operational flight records with scheduled vs. actual times. Used to calibrate the delay distribution model and classify disruption severity.

Field	Type	Description	used_in_code
flight_no	string	Flight number (e.g. `TS793`).	Yes
date	date	Operating date of the flight.	Yes
origin	string	Departure station IATA code.	Yes
destination	string	Arrival station IATA code.	Yes
fleet	string	Aircraft fleet type.	Yes
sched_dep_local	datetime	Scheduled departure in local time.	No
sched_arr_local	datetime	Scheduled arrival in local time.	No
actual_dep_local	datetime	Actual departure in local time.	No
actual_arr_local	datetime	Actual arrival in local time.	No
delay_minutes	integer	Departure delay in minutes (0 = on time). Key field for delay model calibration.	Yes
iata_delay_code	string	IATA standard delay code (e.g. `41` = technical, `93` = crew).	No

Field	Type	Description	used_in_code
cancelled	integer	Cancellation flag: `0` = operated, `1` = cancelled.	Yes
cancel_reason	string	Reason for cancellation, if applicable.	No

D9 — `d9_station_days.csv`

Daily disruption classification per station. One row per station per day, providing the operational severity regime that drives the Markov chain calibration.

Field	Type	Description	used_in_code
station	string	Station IATA code (e.g. `AMS`, `YUL`).	Yes
date	date	Calendar date.	Yes
disruption_class	string	Severity regime: `CALM`, `NORMAL`, `DISRUPTED`, `SEVERE`.	Yes
season	string	Season label: `winter_sun`, `summer_atlantic`, `shoulder`.	Yes
intensity_multiplier	float	Numeric multiplier reflecting disruption intensity (e.g. `0.5` for CALM, `1.0` for NORMAL).	No

D10 — `d10_positioning_invoices.csv`

Positioning cost invoices (deadhead flights, hotels, ground transport) incurred when covering open pairings away from base. Used for backtest cost validation.

Field	Type	Description	used_in_code
date	date	Invoice date.	Yes
station	string	Station where the cost was incurred.	Yes
cost_type	string	Category: `DEADHEAD`, `HOTEL`, `TRANSPORT`.	No
amount_cad	float	Invoice amount in Canadian dollars.	Yes
crew_id	string	Foreign key to D1 — the crew member being positioned.	Yes
pairing_id	string	Foreign key to D3 — the pairing being covered.	Yes

D11 — `d11_training_sessions.csv`

Crew training and simulator session records. One row per session. Currently declared in the schema for future use (drafting tier classification for training disruption).

Field	Type	Description	used_in_code
crew_id	string	Foreign key to D1.	Yes

Field	Type	Description	used_in_code
session_type	string	Training type: `SIM` (simulator) or `GROUND` (classroom).	No
start_date	date	Session start date.	Yes
end_date	date	Session end date.	Yes
requalification_deadline	date	Deadline by which this training must be completed to maintain qualification.	No
reschedule_cost_cad_estimate	float	Estimated cost in CAD to reschedule this session if disrupted by a draft.	No